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**EC-226-CBCS**

**B.E., IV Semester**

Examination, June 2020

**Choice Based Credit System (CBCS)**

**Integrated Circuits and its Applications**

**Time : Three Hours**

**Maximum Marks : 60**

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Define feedback topologies. Explain with diagram.  
b) What is the concept of positive feedback and explain LC oscillator (Any one type).
2. a) Explain block diagram of op-amp and compare ideal and practical characteristics of op-amp.  
b) What is differential amplifier and explain any one type in detail.
3. a) Explain op-amp operation as filter and describe butterworth 2<sup>nd</sup> order low pass filter.  
b) i) Explain phase-shift oscillator.  
ii) Derive gain of non-inverting amplifier.
4. a) Describe series clipper circuit with waveform.  
b) Explain Astable Multivibrator. Using IC-555 timer.
5. a) Compare Fixed and Adjustable voltage regulator.  
b) Write down the characteristics of standard regulator IC's in detail.

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6.
  - a) Describe the amplifier with negative feedback and write its advantages.
  - b) Write down any five characteristics of op-amp in detail.
7.
  - a) Explain antilog amplifier and derive the output voltage expression.
  - b) Discuss sample and hold circuit with diagram.
8. Write short notes on:
  - a) CMRR
  - b) Zero Crossing Detector
  - c) Clamper Circuit (positive peak clamper)

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